ImpactHub

Software Requirements Specification

Version 1.0

March 2026

Prepared by: Aniket

Roll No: 2310990110

Contents

1. Introduction 5
1.1 Purpose 5
1.2 Project Overview 5
1.3 Scope 5
1.3.1 In Scope 5
1.3.2 Out Of Scope 5
1.4 Intended Audience 5
1.5 Definitions and Acronyms 6
1.6 Document Overview 6
2. Project Goals & Stakeholders 7
2.1 Problem Statement 7
2.2 Project Goals 7
2.3 Stakeholders 7
2.4 User Personas 8
2.4.1. NGO Representative 8
2.4.2. Volunteer 8
2.4.3. Platform Admin 8
2.5 Success Criteria 8
3. Functional Requirements 9
3.1 Authentication & Authorization 9
3.2 NGO Management 9
3.3 Event Management 9
3.4 Volunteer Features 10
3.5 Admin Panel 10
3.6 Notifications 10
3.7 Requirements Traceability Summary 11
4. Non-Functional Requirements 12
4.1 Performance 12
4.2 Security 12
4.3 Scalability 12
4.4 Usability 13

4.5 Maintainability 13
4.6 Reliability & Availability 13
4.7 Non-Functional Requirements Summary 14
5. System Architecture & Tech Stack 15
5.1 Architectural Overview 15
5.2 Tech Stack 15
5.3 System Architecture Diagram 16
5.4 API Design 16
5.4.1 Authentication Routes 16
5.4.2 NGO Routes 16
5.4.3 Event Routes 16
5.4.4 Volunteer Routes 16
5.4.5 Admin Routes 17
5.5 Folder Structure 17
5.5.1 Backend 17
5.5.2 Frontend 17
5.6 Deployment Architecture 17
6. Data Models 18
6.1 User 18
6.2 NGO 18
6.3 Event 19
6.4 Relationships Between Collections 19
6.5 Data Validation Rules 19
7. UML Diagrams 21
7.1 Use Case Diagram 21
7.2 Class Diagram 21
7.3 Sequence Diagram 22
7.4 Activity Diagram 23
7.5 System Architecture Diagram 24
8. Deployment Plan & Future Enhancements 25
8.1 Deployment Overview 25
8.2 Deployment Stack 25
8.3 Deployment Process 25
8.3.1 Frontend Deployment — Vercel 25

8.3.2 Backend Deployment — Render 25
8.3.3 Database Deployment — MongoDB Atlas 26
8.4 Environment Variables 26
8.4.1 Backend Environment Variables 26
8.4.2 Frontend Environment Variables 26
8.5 Deployment Checklist 27
8.6 Future Enhancements 27
8.7 Version Roadmap 28

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for ImpactHub, a web-based platform designed to connect Non-Governmental Organizations (NGOs), volunteers, and donors. This document serves as a reference for the design, development, and evaluation of the system.

1.2 Project Overview

ImpactHub is a centralized web-based platform that addresses the problem of fragmented information in the social service sector. It enables NGOs to register, create profiles, and post events and initiatives, while allowing volunteers to discover and participate in social impact activities. The platform is built using the MERN stack and is designed to be scalable, secure, and intuitive for all user roles.

1.3 Scope

1.3.1 In Scope

The following capabilities are included within the scope of this project:

- User registration and login with role-based access control (Admin, NGO, Volunteer)
- NGO profile creation and management
- Event creation, modification, and deletion by NGOs
- Event discovery and registration by volunteers
- Admin panel for NGO verification and content moderation
- Cloud deployment on Vercel, Render, and MongoDB Atlas

1.3.2 Out Of Scope

The following features are considered future enhancements and are not part of the current release:

- Donation and payment integration
- Real-time chat between volunteers and NGOs
- AI-based event recommendations
- NGO analytics and participation trend reports

1.4 Intended Audience

This document is intended for the following audiences:

- **Developers** responsible for building and maintaining the ImpactHub platform
- **Project Evaluators** assessing the system against its requirements and design
- **Stakeholders** seeking an understanding of the system's capabilities and limitations

1.5 Definitions and Acronyms

The following terms and acronyms are used throughout this document:

Term	Definition
SRS	Software Requirements Specification
MERN	MongoDB, Express.js, React.js, Node.js
NGO	Non-Governmental Organization
JWT	JSON Web Token, used for secure authentication
API	Application Programming Interface
CRUD	Create, Read, Update, Delete operations
Admin	Platform administrator with elevated system privileges
Volunteer	A registered user who discovers and joins NGO events
Render	Cloud platform used for backend deployment
Vercel	Cloud platform used for frontend deployment

1.6 Document Overview

The remainder of this document is organized as follows. Section 2 outlines the project goals and stakeholders. Section 3 details the functional requirements. Section 4 covers non-functional requirements. Section 5 describes the system architecture and tech stack. Section 6 presents the data models. Section 7 contains all UML diagrams. Section 8 outlines the deployment plan and future enhancements.

2. Project Goals & Stakeholders

2.1 Problem Statement

The social service sector suffers from a significant lack of centralized information infrastructure. NGOs and non-profit organizations typically rely on fragmented channels such as social media posts, word of mouth, and disconnected websites to communicate their events and initiatives. This results in low volunteer participation, poor donor visibility, and inefficient event management. ImpactHub aims to solve this by providing a single, structured, and accessible platform for all stakeholders in the social impact ecosystem.

2.2 Project Goals

The following goals define the purpose and direction of the ImpactHub platform:

- Provide NGOs with a dedicated space to register, build a verified profile, and manage events efficiently
- Enable volunteers to discover, filter, and register for social impact events based on their interests and location
- Establish a trust layer through admin-based NGO verification to prevent fake or malicious organizations
- Build a scalable and maintainable platform using the MERN stack that can accommodate future enhancements
- Deliver an intuitive and accessible user experience for all roles including non-technical NGO staff and first-time volunteers

2.3 Stakeholders

The following stakeholders interact with or are affected by the ImpactHub platform:

Stakeholder	Role	Primary Interest
NGO/ Non-Profit	Register and manage events, track volunteers	Reach more volunteers and manage participation efficiently
Volunteer	Discover and register for events	Find meaningful opportunities aligned with their interests
Donor	Browse NGOs and contribute to causes	Support verified and trustworthy organizations
Admin	Verify NGOs, moderate content	Maintain platform integrity and trust
Developer	Build and maintain the platform	Deliver a stable, scalable, and secure system
Project Evaluator	Assess the system against requirements	Verify completeness, quality, and functionality

2.4 User Personas

2.4.1. NGO Representative

A representative from a small to mid-sized non-profit organization who manages their organization's digital presence. They are moderately tech-savvy and need a straightforward interface to post events, track registrations, and manage their profile without requiring technical expertise.

2.4.2. Volunteer

A college student or working professional looking to contribute to social causes in their spare time. They want to quickly browse available events, filter by location or category, and register with minimal friction. They may access the platform from both desktop and mobile.

2.4.3. Platform Admin

An internal administrator responsible for maintaining the quality and trustworthiness of the platform. They review and approve NGO registration requests, monitor reported content, and ensure the platform remains free of fraudulent activity.

2.5 Success Criteria

The ImpactHub platform will be considered successful when the following criteria are met:

- NGOs can register, get verified, and post events without requiring technical assistance
- Volunteers can discover and register for events within three clicks from the homepage
- All user roles have clearly separated access and functionality
- The platform is fully deployed and accessible online via cloud infrastructure
- The codebase is modular, readable, and maintainable for future development

3. Functional Requirements

The functional requirements define the specific behaviors and functions the ImpactHub system must support. Each requirement is assigned a unique identifier for traceability.

3.1 Authentication & Authorization

ID	Requirement	Role
FR-01	The system shall allow new users to register using a full name, email address, and password	All
FR-02	The system shall authenticate users via email and password and return a JWT token upon successful login	All
FR-03	The system shall enforce role-based access control with three distinct roles: Admin, NGO, and Volunteer	All
FR-04	The system shall restrict access to protected routes based on the authenticated user's role	All
FR-05	The system shall securely hash all passwords before storing them in the database	All
FR-06	The system shall invalidate sessions upon logout by discarding the JWT token on the client side	All

3.2 NGO Management

ID	Requirement	Role
FR-07	The system shall allow NGOs to create and manage an organizational profile including name, description, address, and contact information	NGO
FR-08	The system shall allow NGOs to update their profile information at any time	NGO
FR-09	The system shall display an NGO's verification status on their public profile	All
FR-10	The system shall allow NGOs to view a list of volunteers registered for their events	NGO
FR-11	The system shall restrict NGO dashboard access to verified NGO accounts only	NGO

3.3 Event Management

ID	Requirement	Role
FR-12	The system shall allow verified NGOs to create new events with a title, description, date, time, and location	NGO
FR-13	The system shall allow NGOs to edit the details of their existing events	NGO
FR-14	The system shall allow NGOs to delete events they have created	NGO
FR-15	The system shall display all active upcoming events on the volunteer-facing event discovery page	Volunteer

FR-16	The system shall allow volunteers to filter events by location and date	Volunteer
FR-17	The system shall display full event details including NGO name, date, time, location, and description on the event detail page	All

3.4 Volunteer Features

ID	Requirement	Role
FR-18	The system shall allow volunteers to register for an event with a single action	Volunteer
FR-19	The system shall prevent a volunteer from registering for the same event more than once	Volunteer
FR-20	The system shall allow volunteers to cancel their registration for an event	Volunteer
FR-21	The system shall display a list of all events a volunteer has registered for on their personal dashboard	Volunteer
FR-22	The system shall send a confirmation notification to the volunteer upon successful event registration	Volunteer

3.5 Admin Panel

ID	Requirement	Role
FR-23	The system shall provide an admin dashboard listing all NGO registration requests with their current status	Admin
FR-24	The system shall allow the admin to approve or reject NGO registration requests	Admin
FR-25	The system shall notify the NGO via a status update when their registration is approved or rejected	Admin
FR-26	The system shall allow the admin to remove events or profiles flagged as fraudulent or inappropriate	Admin
FR-27	The system shall provide the admin with a list of all registered users and their roles	Admin

3.6 Notifications

ID	Requirement	Role
FR-28	The system shall notify volunteers with a confirmation message upon successful event registration	Volunteer
FR-29	The system shall notify NGOs when a volunteer registers for one of their events	NGO
FR-30	The system shall notify NGOs when their account verification status is updated by the admin	NGO

3.7 Requirements Traceability Summary

The table below summarizes all functional requirements by module for quick reference and traceability during development and evaluation.

Module	Requirement IDs	Total
Authentication & Authorization	FR-01 to FR-06	6
NGO Management	FR-07 to FR-11	5
Event Management	FR-12 to FR-17	6
Volunteer Features	FR-18 to FR-22	5
Admin Panel	FR-23 to FR-27	5
Notifications	FR-28 to FR-30	3
Total	**FR-01 to FR-30**	**30**

4. Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the ImpactHub system. These requirements govern how the system performs its functions rather than what functions it performs.

4.1 Performance

ID	Requirement	Metric
NFR-01	The system shall load the homepage and event listing page within an acceptable response time under normal network conditions	Page load under 3 seconds
NFR-02	The backend API shall respond to standard requests without significant delay	API response under 500ms
NFR-03	The system shall support multiple concurrent users without degradation in performance	Minimum 100 concurrent users
NFR-04	Database queries shall be optimized using appropriate indexing on frequently searched fields such as event date and location	Query response under 200ms

4.2 Security

ID	Requirement	Metric
NFR-05	All user passwords shall be hashed before storage using a recognized hashing algorithm	bcrypt with salt rounds
NFR-06	All protected API routes shall require a valid JWT token for access	JWT expiry set to 7 days
NFR-07	The system shall validate and sanitize all user inputs on both the client and server side to prevent injection attacks	Input validation on all forms
NFR-08	Sensitive configuration values such as database credentials and JWT secrets shall be stored as environment variables and never hardcoded	.env file, excluded from version control
NFR-09	The system shall enforce HTTPS for all client-server communication in the deployed environment	SSL via Vercel and Render

4.3 Scalability

ID	Requirement	Metric
NFR-10	The system architecture shall be modular to allow independent scaling of the frontend and backend	Decoupled MERN architecture
NFR-11	The database shall be hosted on MongoDB Atlas to leverage cloud-native horizontal scaling capabilities	MongoDB Atlas free to paid tier

NFR-12	The codebase shall follow a modular folder structure separating routes, controllers, models, and middleware	MVC pattern on backend

4.4 Usability

ID	Requirement	Metric
NFR-13	The user interface shall be intuitive enough for non-technical users such as NGO staff and first-time volunteers to navigate without training	Usability tested with target personas
NFR-14	The platform shall be fully responsive and functional on both desktop and mobile screen sizes	Responsive design using CSS/Tailwind
NFR-15	All error messages displayed to the user shall be clear, descriptive, and actionable	No raw error codes exposed to users
NFR-16	The volunteer event registration process shall be completable within three clicks from the homepage	Verified via user flow testing

4.5 Maintainability

ID	Requirement	Metric
NFR-17	The codebase shall use TypeScript across both frontend and backend for type safety and maintainability	Strict mode enabled via tsconfig.json
NFR-18	The backend shall follow the MVC architectural pattern separating models, routes, and controllers	Express MVC structure
NFR-19	The frontend shall be built using reusable React components to minimize code duplication	Component-based architecture
NFR-20	All source code shall be version controlled using Git and hosted on a public GitHub repository	GitHub repository

4.6 Reliability & Availability

ID	Requirement	Metric
NFR-21	The deployed application shall maintain high availability during the evaluation period	Hosted on Vercel and Render
NFR-22	The system shall handle unexpected errors gracefully without crashing and return appropriate error responses to the client	Global error handler on backend
NFR-23	The database shall be backed up automatically through MongoDB Atlas cloud infrastructure	MongoDB Atlas automated backups

4.7 Non-Functional Requirements Summary

The table below summarizes all non-functional requirements by category for quick reference.

Category	Requirement IDs	Total
Performance	NFR-01 to NFR-04	4
Security	NFR-05 to NFR-09	5
Scalability	NFR-10 to NFR-12	3
Usability	NFR-12 to NFR-16	4
Maintainability	NFR-17 to NFR-20	4
Reliability & Availability	NFR-21 to NFR-23	3
Total	**NFR-01 to NFR-23**	**23**

5. System Architecture & Tech Stack

This section describes the overall architecture of the ImpactHub platform, the technologies used at each layer, and the rationale behind key technical decisions.

5.1 Architectural Overview

ImpactHub follows a three-tier client-server architecture consisting of a presentation layer, an application logic layer, and a data layer. The frontend and backend are fully decoupled, communicating exclusively through a RESTful API. This separation allows each layer to be developed, deployed, and scaled independently.

The three tiers are organized as follows:

- **Presentation Layer** — The React.js or Next.js frontend served via Vercel, responsible for rendering the user interface and handling client-side interactions
- **Application Layer** — The Node.js and Express.js backend hosted on Render, responsible for business logic, authentication, and API endpoints
- **Data Layer** — The MongoDB database hosted on MongoDB Atlas, responsible for persistent storage of all application data

5.2 Tech Stack

The following technologies form the core stack of the ImpactHub platform:

Layer	Technology	Version	Purpose
Language	TypeScript	5.x	Statically typed superset of JavaScript used across the backend
Frontend	React.js/Next.js	14.x	User interface with built-in TypeScript support
Backend	Node.js	20.x	Server-side runtime environment
Backend Framework	Express.js	4.x	RESTful API development and routing
Database	MongoDB	7.x	NoSQL document-based data storage
ODM	Mongoose	8.x	MongoDB object modelling for Node.js
Authentication	JSON Web Token (JWT)	-	Stateless user authentication
Password Hashing	bcrypt	-	Secure password storage
Frontend Deployment	Vercel	-	Cloud hosting for the frontend application
Backend Deployment	Render	-	Cloud hosting for the backend API
Database Hosting	MongoDB Atlas	-	Managed cloud database service
Version Control	Git + Github	-	Source code management and collaboration

5.3 System Architecture Diagram

The diagram below illustrates the high-level architecture of the ImpactHub platform, showing the interaction between the client, server, and database layers across the deployment environment.

5.4 API Design

The ImpactHub backend exposes a RESTful API consumed by the frontend. All API routes are prefixed with /api. The following table outlines the core endpoints grouped by module:

5.4.1 Authentication Routes

Method	Endpoint	Description	Access
POST	/api/auth/register	Register a new user	Public
POST	/api/auth/login	Authenticate user and return JWT	Public
GET	/api/auth/me	Get current authenticated user profile	Private

5.4.2 NGO Routes

Method	Endpoint	Description	Access
GET	/api/ngos	Get all verified NGOs	Public
GET	/api/ngos/:id	Get a single NGO profile by ID	Public
POST	/api/ngos	Create a new NGO profile	NGO
PUT	/api/ngos/:id	Update an existing NGO profile	NGO
DELETE	/api/ngos/:id	Delete an NGO profile	Admin

5.4.3 Event Routes

Method	Endpoint	Description	Access
GET	/api/events	Get all upcoming events	Public
GET	/api/events/:id	Get a single event by ID	Public
POST	/api/events	Create a new event	NGO
PUT	/api/events/:id	Update an existing event	NGO
DELETE	/api/events/:id	Delete an event	NGO, Admin

5.4.4 Volunteer Routes

Method	Endpoint	Description	Access
POST	/api/events/:id/register	Register for an event	Volunteer
DELETE	/api/events/:id/register	Cancel event registration	Volunteer
GET	/api/volunteers/events	Get all events registered by volunteer	Volunteer

5.4.5 Admin Routes

Method	Endpoint	Description	Access
GET	/api/admin/ngos	Get all NGO registration requests	Admin
PUT	/api/admin/ngos/:id/verify	Approve or reject an NGO	Admin
GET	/api/admin/users	Get all registered users	Admin
DELETE	/api/admin/users/:id	Remove a user from the platform	Admin

5.5 Folder Structure

The project follows a clean and modular folder structure to ensure maintainability and separation of concerns. The high-level structure is as follows:

5.5.1 Backend

```
backend/
├── src/
│   ├── config/          # Database connection and env config
│   ├── controllers/     # Business logic for each route module
│   ├── middleware/      # Authentication and RBAC middleware
│   ├── models/          # Mongoose schemas
│   ├── routes/          # Express route definitions
│   ├── types/           # Shared TypeScript interfaces and types
│   └── utils/           # Helper functions and utilities
├── dist/                # Compiled JavaScript output
├── .env                 # Environment variables
├── tsconfig.json        # TypeScript compiler configuration
└── server.ts            # Application entry point
```

5.5.2 Frontend

```
frontend/
├── public/          # Static assets
├── src/
│   ├── components/  # Reusable UI components
│   ├── pages/       # Page-level components or Next.js pages
│   ├── context/     # Auth context and global state
│   ├── hooks/       # Custom React hooks
│   ├── services/    # API call functions
│   └── utils/       # Frontend helper functions
└── package.json
```

5.6 Deployment Architecture

The ImpactHub platform is deployed across three cloud services as outlined below:

Layer	Service	URL Pattern	Notes
Frontend	Vercel	impacthub.vercel.app	Auto-deploys on push to main branch
Backend	Render	impacthub-api.onrender.com	Always-on web service
Database	MongoDB Atlas	Cloud hosted cluster	Automated backups enabled

6. Data Models

This section describes the data models used in the ImpactHub platform. The application uses MongoDB as its database, with Mongoose as the ODM layer to define and enforce schemas. The three core collections are User, NGO, and Event.

6.1 User

The User collection stores all registered users across all roles including Admin, NGO representatives, and Volunteers. Role-based access control is enforced based on the role field.

Field	Type	Required	Description
_id	ObjectId	Auto	Unique identifier generated by MongoDB
name	String	Yes	Full name of the user
email	String	Yes	Unique email address used for login
password	String	Yes	Bcrypt hashed password
role	String (Enum)	Yes	One of: Admin, NGO, Volunteer
registeredEvents	Array of ObjectId	No	References to Event documents the volunteer has joined
createdAt	Date	Auto	Timestamp of account creation
updatedAt	Date	Auto	Timestamp of last profile update

6.2 NGO

The NGO collection stores organizational profiles created by users with the NGO role. Each NGO must be verified by an Admin before they can post events or appear in public listings.

Field	Type	Required	Description
_id	ObjectId	Auto	Unique identifier generated by MongoDB
user	ObjectId (ref: User)	Yes	Reference to the User account of the NGO representative
name	String	Yes	Official name of the NGO
description	String	Yes	Brief description of the NGO and its mission
address	String	Yes	Physical address of the organization
contactEmail	String	Yes	Public contact email for the NGO
contactPhone	String	Yes	Public contact phone number
verified	Boolean	Yes	Verification status set by Admin, defaults to false
events	Array of ObjectId	No	References to Event documents created by this NGO
createdAt	Date	Auto	Timestamp of profile creation

updatedAt	Date	Auto	Timestamp of last profile update

6.3 Event

The Event collection stores all events created by verified NGOs. Each event tracks its participants and maintains a reference to the NGO that created it.

Field	Type	Required	Description
_id	ObjectId	Auto	Unique identifier generated by MongoDB
title	String	Yes	Title of the event
description	String	Yes	Detailed description of the event and its purpose
ngo	ObjectId (ref: NGO)	Yes	Reference to the NGO that created the event
date	Date	Yes	Scheduled date of the event
time	String	Yes	Scheduled time of the event
location	String	Yes	Physical location or venue of the event
participants	Array of ObjectId	No	References to User documents of registered volunteers
status	String (Enum)	Yes	One of: Upcoming, Ongoing, Completed, Cancelled
createdAt	Date	Auto	Timestamp of event creation
updatedAt	Date	Auto	Timestamp of last event update

6.4 Relationships Between Collections

The following describes the relationships between the three core collections:

- A **User** can have one **NGO** profile if their role is NGO
- A **User** can register for many **Events** — stored as an array of ObjectIds in the User document
- An **NGO** can create many **Events** — stored as an array of ObjectIds in the NGO document
- An **Event** belongs to one **NGO** and can have many **Users** as participants

Relationship	Type	Description
User → NGO	One to One	One NGO user account maps to one NGO profile
NGO → Event	One to Many	One NGO can create multiple events
User → Event	Many to Many	Many volunteers can register for many events

6.5 Data Validation Rules

The following validation rules are enforced at the Mongoose schema level before any data is persisted to the database:

Field	Rule	Error Message
User.email	Must be unique and match valid email format	Email already in use or invalid format
User.password	Minimum 8 characters	Password must be at least 8 characters
User.role	Must be one of: Admin, NGO, Volunteer	Invalid role specified
NGO.verified	Defaults to false on creation	-
Event.status	Must be one of: Upcoming, Ongoing, Completed, Cancelled	Invalid status value
Event.date	Must be a valid future date at time of creation	Event date must be in the future

7. UML Diagrams

This section presents the UML diagrams for the ImpactHub platform. These diagrams collectively describe the system's actors, data structures, interactions, workflows, and deployment architecture.

7.1 Use Case Diagram

The Use Case Diagram illustrates the interactions between the three primary actors — Admin, NGO, and Volunteer — and the core functionalities of the ImpactHub system. Each use case represents a distinct action a user can perform within the platform.

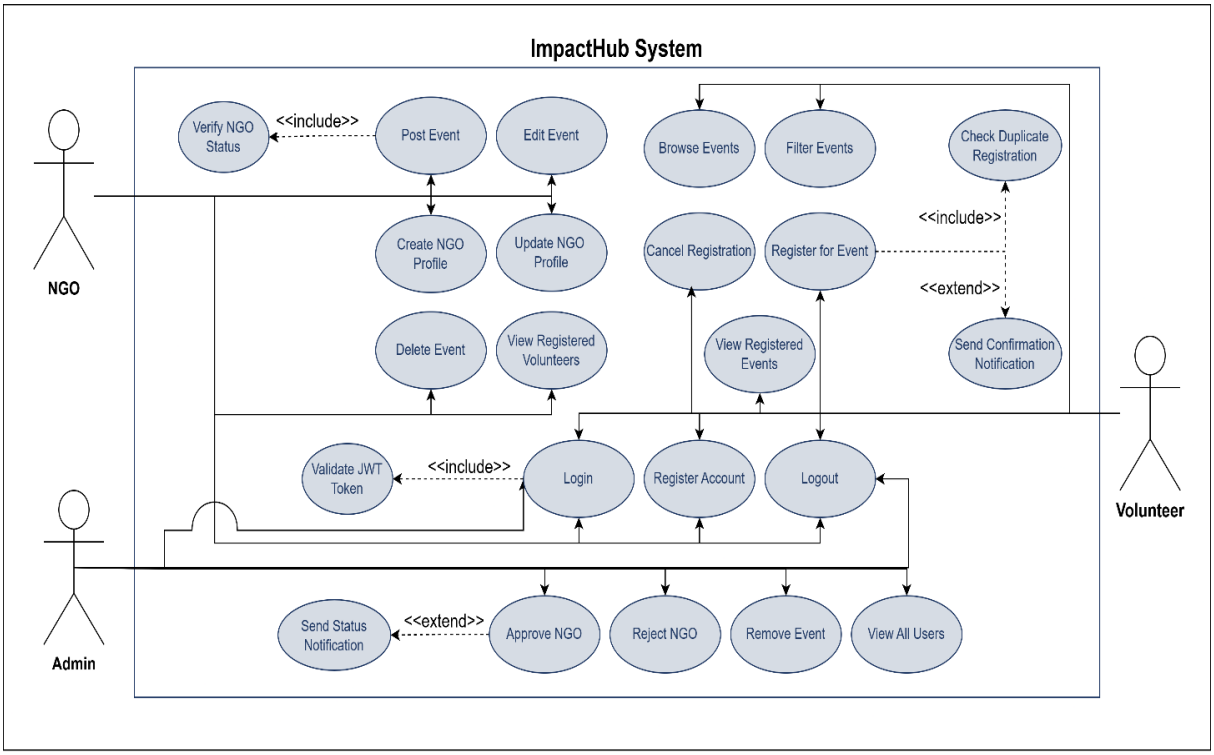

Figure 1 — ImpactHub Use Case Diagram

7.2 Class Diagram

The Class Diagram represents the three core data entities of the ImpactHub platform — User, NGO, and Event — along with their attributes, data types, and the relationships between them. This diagram directly maps to the MongoDB collections and Mongoose schemas defined in Section 6.

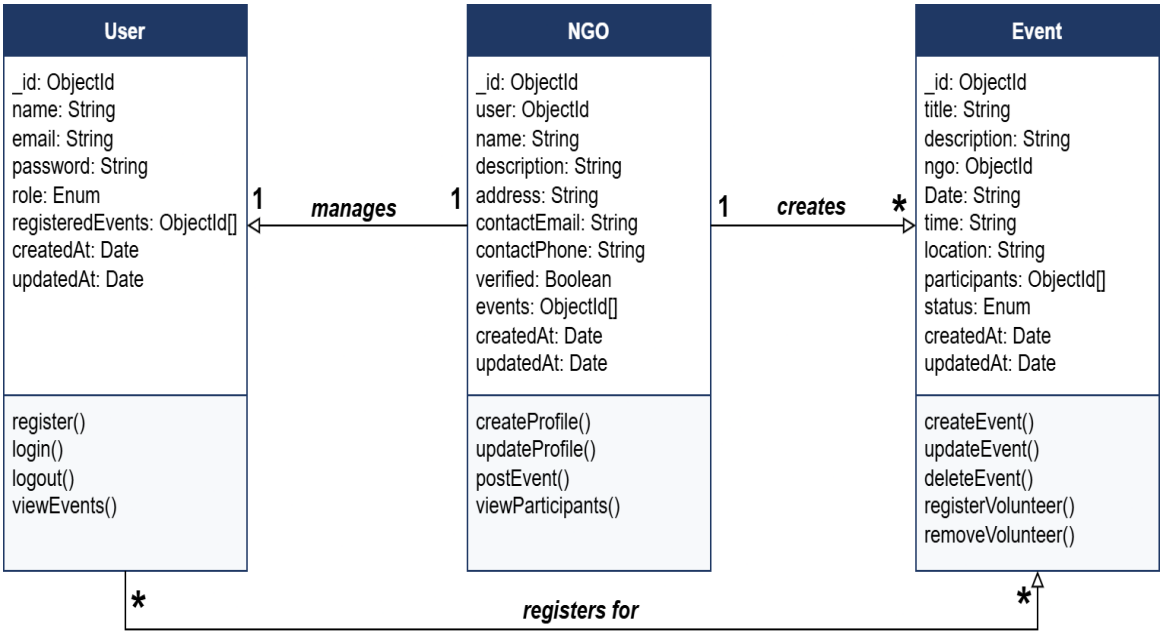

Figure 2 — ImpactHub Class Diagram

7.3 Sequence Diagram

The Sequence Diagram illustrates the step-by-step interaction between system components during the process of a volunteer registering for an event. It shows the order of messages exchanged between the user interface, backend API, authentication middleware, and database.

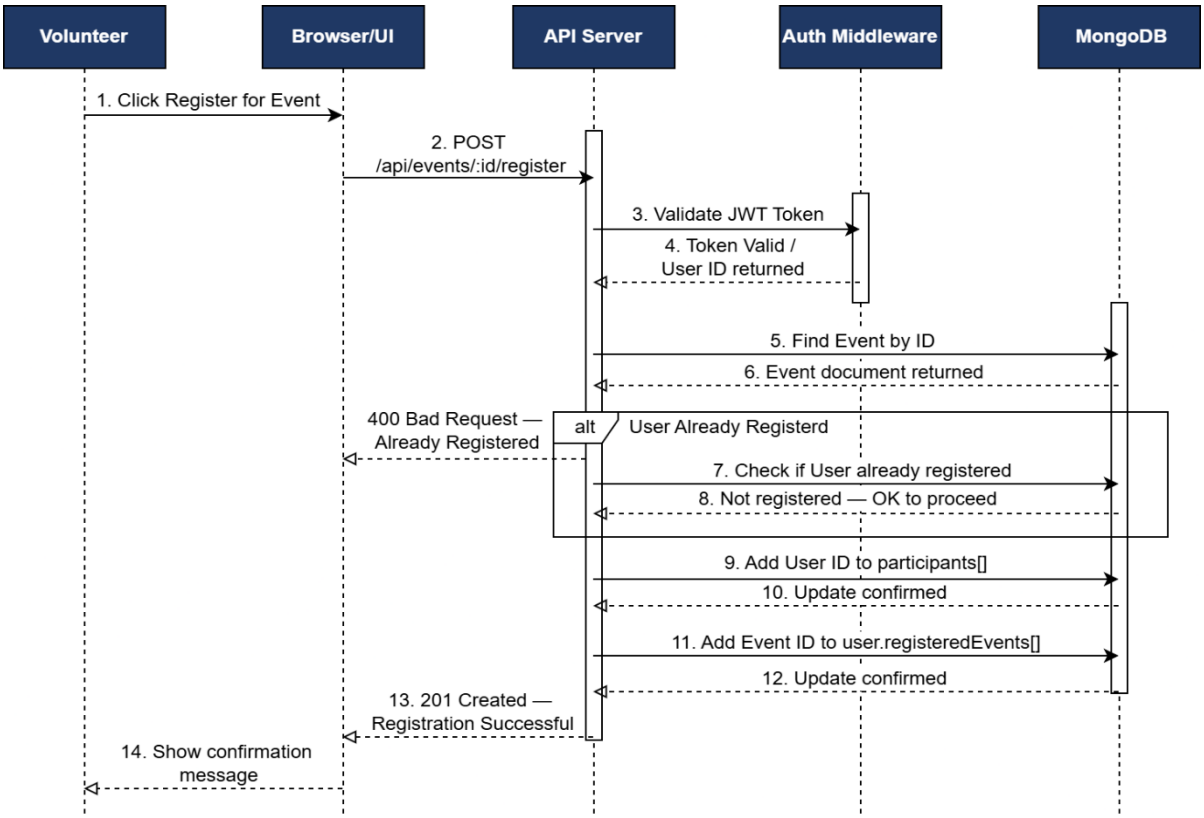

Figure 3 — Volunteer Event Registration Sequence Diagram

7.4 Activity Diagram

The Activity Diagram illustrates the complete workflow of a volunteer's journey through the ImpactHub platform, from initial login through to event registration confirmation. It captures all decision points, alternate paths, and system responses within the flow.

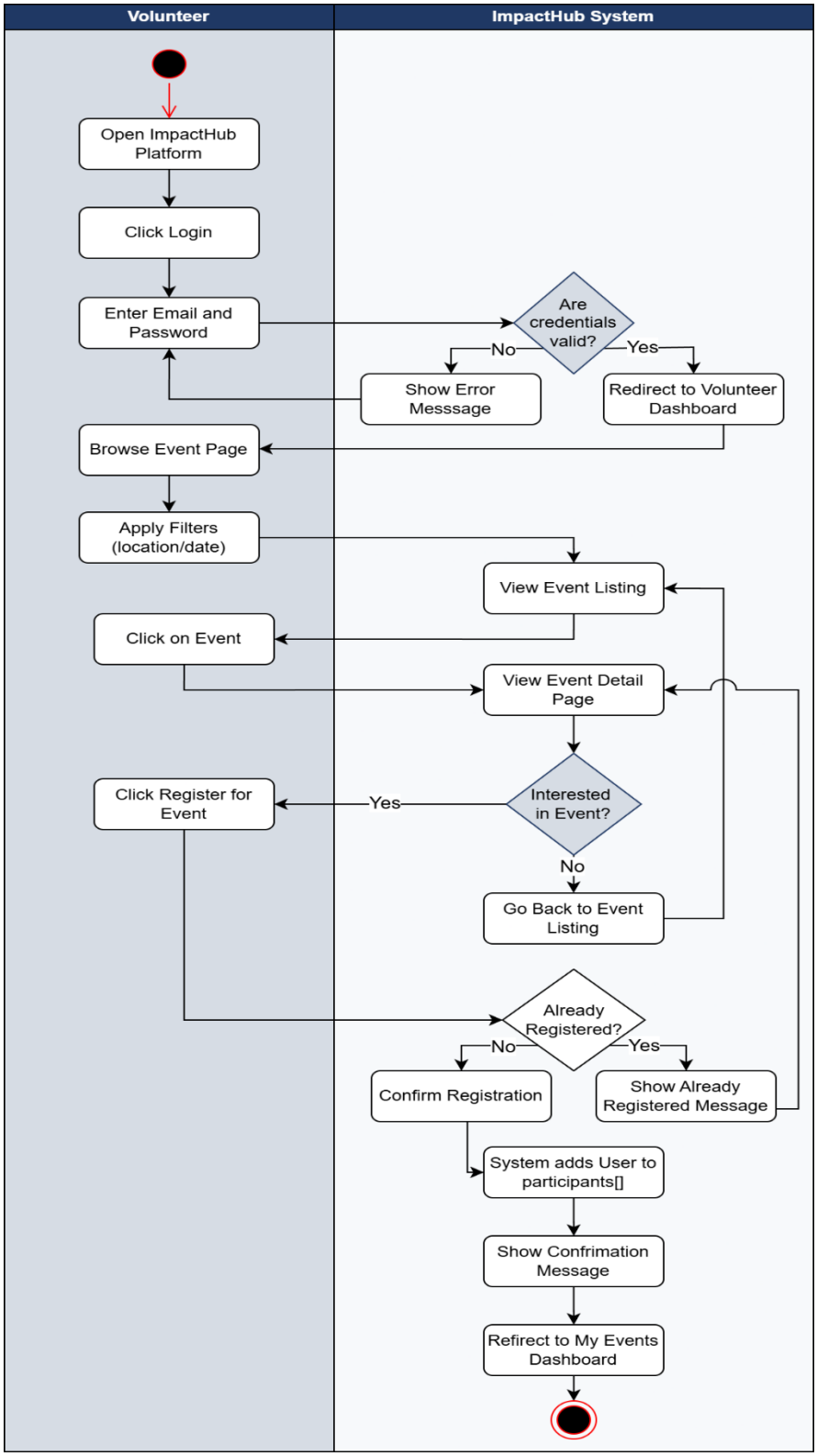

Figure 4 — Volunteer Journey Activity Diagram

7.5 System Architecture Diagram

The System Architecture Diagram illustrates the high-level structure of the ImpactHub platform across its three tiers — presentation, application, and data. It shows how the client, backend API, and database interact with each other across the cloud deployment environment.

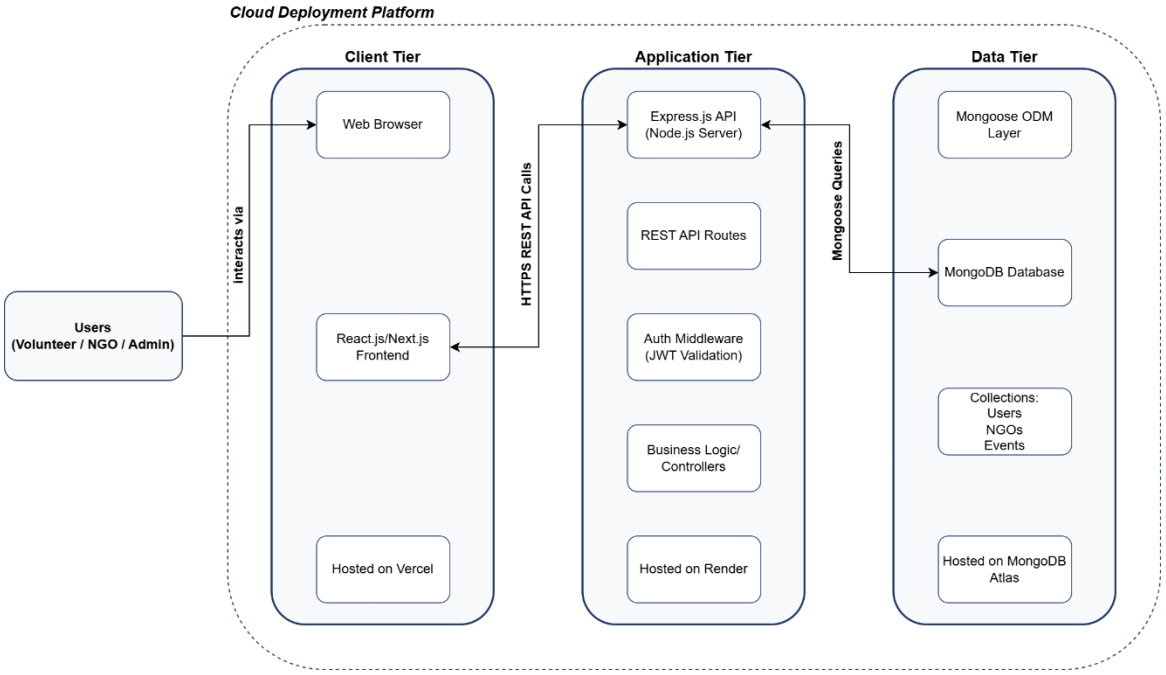

Figure 5 — ImpactHub System Architecture Diagram

8. Deployment Plan & Future Enhancements

This section outlines the deployment strategy for the ImpactHub platform across cloud infrastructure and describes the planned enhancements for future releases beyond the current project scope.

8.1 Deployment Overview

The ImpactHub platform follows a decoupled deployment architecture where the frontend, backend, and database are hosted independently across three separate cloud services. This approach allows each layer to be scaled, updated, and maintained without affecting the others.

8.2 Deployment Stack

Layer	Service	URL Pattern	Details
Frontend	Vercel	impacthub.vercel.app	Auto-deploys on every push to main branch on GitHub
Backend	Render	impacthub-api.onrender.com	Always-on web service, auto-deploys from GitHub
Database	MongoDB Atlas	Cloud hosted cluster	Free tier M0 cluster, automated backups enabled

8.3 Deployment Process

8.3.1 Frontend Deployment — Vercel

The frontend is deployed on Vercel using the following process:

- Connect the GitHub repository to Vercel via the Vercel dashboard
- Set the root directory to /frontend
- Configure environment variables including the backend API base URL
- Every push to the main branch triggers an automatic redeployment
- Vercel provides HTTPS, CDN distribution, and zero-downtime deployments by default

8.3.2 Backend Deployment — Render

The backend API is deployed on Render using the following process:

- Connect the GitHub repository to Render via the Render dashboard
- Set the root directory to /backend
- Set the start command to node server.js
- Configure all environment variables including MONGO_URI, JWT_SECRET, and PORT
- Every push to the main branch triggers an automatic redeployment
- Render provides HTTPS and always-on hosting on the free tier

8.3.3 Database Deployment — MongoDB Atlas

The MongoDB database is hosted on MongoDB Atlas using the following process:

- Create a free tier M0 cluster on MongoDB Atlas
- Whitelist all IP addresses (0.0.0.0/0) to allow connections from Render
- Create a database user with read and write privileges
- Copy the connection string and set it as MONGO_URI in the backend environment variables
- Enable automated backups through the Atlas dashboard

8.4 Environment Variables

The following environment variables are required for the ImpactHub platform to function correctly in the deployed environment. These values are never hardcoded in the codebase and are excluded from version control via .gitignore.

8.4.1 Backend Environment Variables

Variable	Description	Example
MONGO_URI	MongoDB Atlas connection string	mongodb+srv://user:pass@cluster.mongodb.net/impacthub
JWT_SECRET	Secret key used to sign JWT tokens	a_long_random_secret_string
PORT	Port number for the Express server	5000
NODE_ENV	Environment mode	production

8.4.2 Frontend Environment Variables

Variable	Description	Example
NEXT_PUBLIC_API_URL	Base URL of the deployed backend API	https://impacthub-api.onrender.com

8.5 Deployment Checklist

The following checklist must be completed before the platform is considered fully deployed and ready for evaluation:

#	Task	Status
1	MongoDB Atlas cluster created and configured	☐
2	Backend deployed and accessible on Render	☐
3	Frontend deployed and accessible on Vercel	☐
4	All environment variables configured on both services	☐
5	Frontend successfully communicating with backend API	☐
6	User registration and login tested on deployed version	☐
7	NGO profile creation tested on deployed version	☐
8	Event creation and registration tested on deployed version	☐
9	Admin panel tested on deployed version	☐
10	GitHub repository is public with complete code	☐

8.6 Future Enhancements

The following features are planned for future releases of ImpactHub beyond the current project scope. These enhancements are prioritized based on their potential impact on the platform's usability and reach.

#	Feature	Description	Priority
1	Donation Integration	Allow donors to contribute directly to NGOs through the platform using a payment gateway such as Razorpay or Stripe	High
2	Real-Time Chat	Enable real-time messaging between volunteers and NGO representatives using WebSockets or Socket.io	High
3	AI Event Recommendations	Recommend events to volunteers based on their registration history, interests, and location using a machine learning model	Medium
4	NGO Analytics Dashboard	Provide NGOs with visual reports on volunteer participation trends, event performance, and engagement metrics	Medium
5	Mobile Application	Develop a React Native mobile application for iOS and Android to extend platform accessibility	Medium
6	Email Notifications	Send automated email reminders to volunteers before their registered events using a service such as Nodemailer or SendGrid	Low
7	Social Login	Allow users to register and login using Google or LinkedIn OAuth for a faster onboarding experience	Low
8	Volunteer Badges	Introduce a gamification system where volunteers earn badges and recognition based on their participation history	Low

8.7 Version Roadmap

Version	Scope	Target	Status
v1.0	Core platform — authentication, NGO profiles, event management, volunteer registration, admin panel	April 2026	In Progress
v1.1	Email notifications, social login, volunteer badges	Q3 2026	Planned
v2.0	Donation integration, real-time chat, mobile application	Q1 2027	Planned
v3.0	AI recommendations, NGO analytics dashboard	Q3 2027	Planned